1. Which of the following is true about JavaFX?

a) It's primarily used for server-side programming.

b) It's a scripting language.

c) It's a set of Java libraries for building rich client applications.

d) It's designed for data analysis.

2. What class should you extend when creating a JavaFX application?

a) `java.awt. Application`

b) `javax.swing.JFrame`

c) `javafx.application.Application`

d) `javafx.ui.Application`

3. Which layout manager is used to arrange UI components in a horizontal row?

a) `VBox`

b) `HBox`

c) `BorderPane`

d) `FlowPane`

4. Which of the following is used for handling events in JavaFX?

a) `ActionListener`

b) `EventHandler`

c) `MouseListener`

d) `ClickListener`

5. Which method is used to set an action handler for a button in JavaFX?

a) `setOnMouseClick()`

b) `setOnAction()`

c) `setOnClick()`

d) `addActionListener()`

6. Which file format is commonly used to define UI layouts in JavaFX?

a) XML

b) JSON

c) YAML

d) HTML

7. Which CSS pseudo-class is used to style the focused state of a text field in JavaFX?

a) `:focus`

b) `:hover`

c) `:active`

d) `:selected`

8. Which class is used for playing audio and video files in JavaFX?

a) `javafx.media.MediaPlayer`

b) `javafx.scene.media.Media`

c) `javafx.scene.media.AudioPlayer`

d) `javafx.media.AudioClip`

9. What is the purpose of the Scene Graph in JavaFX?

a) To manage database connections

b) To manage the rendering and layout of UI components

c) To handle user authentication

d) To execute background tasks asynchronously

10. Which method is used to start a JavaFX application?

a) `run()`

b) `init()`

c) `start()`

d) `launch()`

11. Which of the following is not a layout manager in JavaFX?

a) `GridPane`

b) `StackPane`

c) `LinePane`

d) `AnchorPane`

12. What is the role of the `BorderPane` layout manager in JavaFX?

a) It arranges UI components in a grid.

b) It stacks UI components on top of each other.

c) It organizes UI components in a horizontal row.

d) It divides the scene into regions: top, bottom, left, right, and center.

13. Which class is used to represent a container for UI components in JavaFX?

a) `javafx.scene.Node`

b) `javafx.scene.Scene`

c) `javafx.scene.layout.Pane`

d) `javafx.scene.control.Control`

14. How do you set the background color of a JavaFX application using CSS?

a) `background-color: red;`

b) `bgcolor: red;`

c) `color: red;`

d) `setBackgroundColor(red);`

15. Which method is used to apply CSS styles to a JavaFX node?

a) `setStylesheet()`

b) `applyStylesheet()`

c) `setStyle()`

d) `addStylesheet()`

16. Which of the following is a valid way to handle a mouse click event in JavaFX?

a) `onMouseClicked()`

b) `setOnMouseClicked()`

c) `addMouseClickedListener()`

d) `onClick()`

17. How do you create a new FXML file in JavaFX Scene Builder?

a) File > New JavaFX File

b) File > New FXML Document

c) File > New Scene

d) File > New Project

18. What is the purpose of the `FlowPane` layout manager in JavaFX?

a) It arranges UI components in a grid.

b) It organizes UI components in a horizontal row.

c) It stacks UI components on top of each other.

d) It divides the scene into regions: top, bottom, left, right, and center.

19. Which method is used to add a UI component to a JavaFX scene?

a) `add()`

b) `addChild()`

c) `addElement()`

d) `getChildren().add()`

20. How do you specify the font size of a JavaFX node using CSS?

a) `font-size: 12pt;`

b) `fontsize: 12pt;`

c) `text-size: 12pt;`

d) `setFontSize(12pt);`

21. Which of the following is not a valid event type in JavaFX?

a) `KeyEvent`

b) `MouseEvent`

c) `TouchEvent`

d) `ActionEvent`

22. Which method is used to remove a UI component from a JavaFX scene?

a) `remove()`

b) `removeChild()`

c) `removeElement()`

d) `getChildren().remove()`

23. How do you set the alignment of a UI component in a `BorderPane` layout in JavaFX?

a) By using the `setAlignment()` method

b) By specifying it in the FXML file

c) By setting the alignment constraint during layout

d) By adding it to the correct region of the `BorderPane`

24. Which of the following is true about FXML in JavaFX?

a) It stands for "Fast XML."

b) It's a programming language.

c) It's used for defining UI layouts in JavaFX.

d) It's only supported in older versions of JavaFX.

25. What is the purpose of the `GridPane` layout manager in JavaFX?

a) It arranges UI components in a vertical column.

b) It organizes UI components in a horizontal row.

c) It arranges UI components in a grid of rows and columns.

d) It stacks UI components on top of each other.

26. Which CSS property is used to specify the padding of a JavaFX node?

a) `padding`

b) `margin`

c) `border`

d) `spacing`

27. How do you set the minimum width of a JavaFX node using CSS?

a) `min-width: 100px;`

b) `minimum-width: 100px;`

c) `width-min: 100px;`

d) `setMinWidth(100px);`

28. Which method is used to set the background image of a JavaFX node using CSS?

a) `setBackgroundImage()`

b) `setStyle()`

c) `setBackground()`

d) `setBackgroundColor()`

29. What is the purpose of the `StackPane` layout manager in JavaFX?

a) It arranges UI components in a vertical column.

b) It organizes UI components in a horizontal row.

c) It stacks UI components on top of each other.

d) It arranges UI components in a grid of rows and columns.

30. How do you set the opacity of a JavaFX node using CSS?

a) `opacity: 0.5;`

b) `transparency: 0.5;`

c) `visible: false;`

d) `setOpacity(0.5);`

31. What is the purpose of the `AnchorPane` layout manager in JavaFX?

a) It arranges UI components in a grid of rows and columns.

b) It stacks UI components on top of each other.

c) It organizes UI components in a horizontal row.

d) It allows you to anchor UI components to the edges of the container.

32. Which method is used to set the border of a JavaFX node using CSS?

a) `setBorder()`

b) `setBorderStyle()`

c) `setBorderWidth()`

d) `setStyle()`

33. What is the purpose of the `Ellipse` class in JavaFX?

a) To draw circles and ovals

b) To represent a rectangle

c) To display text

d) To create lines and shapes

34. How do you set the stroke color of a JavaFX shape using CSS?

a) `stroke-color: red;`

b) `stroke: red;`

c) `line-color: red;`

d) `setStrokeColor(red);`

35. Which method is used to set the rotation angle of a JavaFX node using CSS?

a) `setRotate()`

b) `setRotation()`

c) `rotate()`

d) `transform()`

36. What is the purpose of the `Arc` class in JavaFX?

a) To draw arcs

b) To represent a circle

c) To create rectangles

d) To display text

37. How do you set the fill color of a JavaFX shape using CSS?

a) `fill-color: red;`

b) `background-color: red;`

c) `color: red;`

d) `setFillColor(red);`

38. Which method is used to set the font family of a JavaFX node using CSS?

a) `font-family: Arial;`

b) `font: Arial;`

c) `setFontFamily(Arial);`

d) `setFont(Arial);`

39. What is the purpose of the `Rectangle` class in JavaFX?

a) To draw rectangles

b) To represent a circle

c) To create lines and shapes

d) To display text

40. How do you set the border radius of a JavaFX node using CSS?

a) `border-radius: 5px;`

b) `rounded-corners: 5px;`

c) `corner-radius: 5px;`

d) `setBorderRadius(5px);`

41. What is the purpose of the `Circle` class in JavaFX?

a) To draw circles

b) To represent a rectangle

c) To create lines and shapes

d) To display text

42. Which method is used to set the text color of a JavaFX node using CSS?

a) `text-color: red;`

b) `color: red;`

c) `font-color: red;`

d) `setTextColor(red);`

43. What is the purpose of the `Line` class in JavaFX?

a) To draw lines

b) To represent a rectangle

c) To create circles

d) To display text

44. How do you set the stroke width of a JavaFX shape using CSS?

a) `stroke-width: 2px;`

b) `line-width: 2px;`

c) `border-width: 2px;`

d) `setStrokeWidth(2px);`

45. Which method is used to set the text content of a JavaFX node using CSS?

a) `setTextContent()`

b) `textContent()`

c) `setText()`

d) `setInnerText()`

46. What is the purpose of the `Polygon` class in JavaFX?

a) To draw polygons

b) To represent a rectangle

c) To create circles

d) To display text

47. How do you set the stroke pattern of a JavaFX shape using CSS?

a) `stroke-pattern: dashed;`

b) `stroke-dash-array: dashed;`

c) `line-style: dashed;`

d) `setStrokePattern(dashed);`

48. Which method is used to set the line cap of a JavaFX shape using CSS?

a) `line-cap: round;`

b) `stroke-cap: round;`

c) `line-style: round;`

d) `setLineCap(round);`

49. What is the purpose of the `Text` class in JavaFX?

a) To display text

b) To draw circles

c) To create rectangles

d) To represent a line

50. Which method is used to set the text alignment of a JavaFX node using CSS?

a) `text-align: center;`

b) `align-text: center;`

c) `alignment: center;`

d) `setTextAlignment(center);`

Answers:

1. c) It's a set of Java libraries for building rich client applications.
2. c) `javafx.application.Application`
3. b) `HBox`
4. b) `EventHandler`
5. b) `setOnAction()`
6. a) XML
7. a) `:focus`
8. a) `javafx.media.MediaPlayer`
9. b) To manage the rendering and layout of UI components
10. d) `launch()`
11. c) `LinePane`
12. d) It divides the scene into regions: top, bottom, left, right, and center.
13. c) `javafx.scene.layout.Pane`
14. a) `background-color: red;`
15. c) `setStyle()`
16. b) `setOnMouseClicked()`
17. b) File > New FXML Document
18. d) It arranges UI components in a grid of rows and columns.
19. d) `getChildren().add()`
20. a) `font-size: 12pt;`
21. d) `ActionEvent`
22. d) `getChildren().remove()`
23. d) By adding it to the correct region of the `BorderPane`
24. c) It's used for defining UI layouts in JavaFX.

25. c) It arranges UI components in a grid of rows and columns.

26. a) `padding`

27. a) `min-width: 100px;`

28. b) `setStyle()`

29. c) It stacks UI components on top of each other.

30. a) `opacity: 0.5;`

31. d) It allows you to anchor UI components to the edges of the container.

32. a) `setBorder()`

33. a) To draw circles and ovals

34. b) `stroke: red;`

35. a) `setRotate()`

36. a) To draw arcs

37. a) `fill-color: red;`

38. a) `font-family: Arial;`

39. a) To draw rectangles

40. a) `border-radius: 5px;`

41. a) To draw circles

42. b) `color: red;`

43. a) To draw lines

44. a) `stroke-width: 2px;`

45. c) `setText()`

46. a) To draw polygons

47. b) `stroke-dash-array: dashed;`

48. b) `stroke-cap: round;`

49. a) To display text

50. a) `text-align: center;`